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# Adaptive optics via pupil segmentation for high-resolution imaging in biological tissues

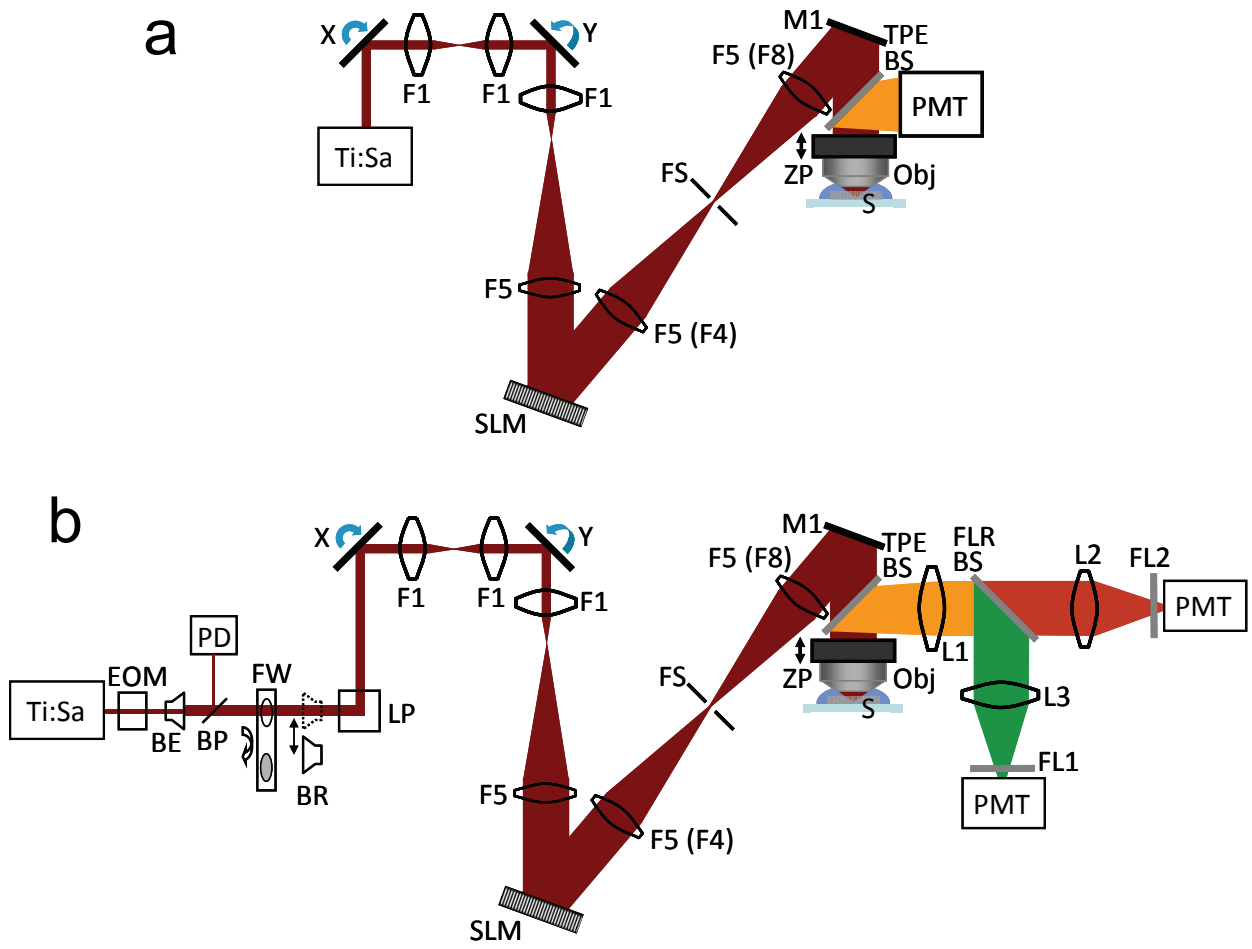
Na Ji, Daniel E Milkie & Eric Betzig

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*Note: SupplementaryVideos 1–3 are available on the Nature Methods website.*

**Supplementary Figure 1** Simplified and detailed schematics for our adaptive-optical (AO) two-photon fluorescence microscope.

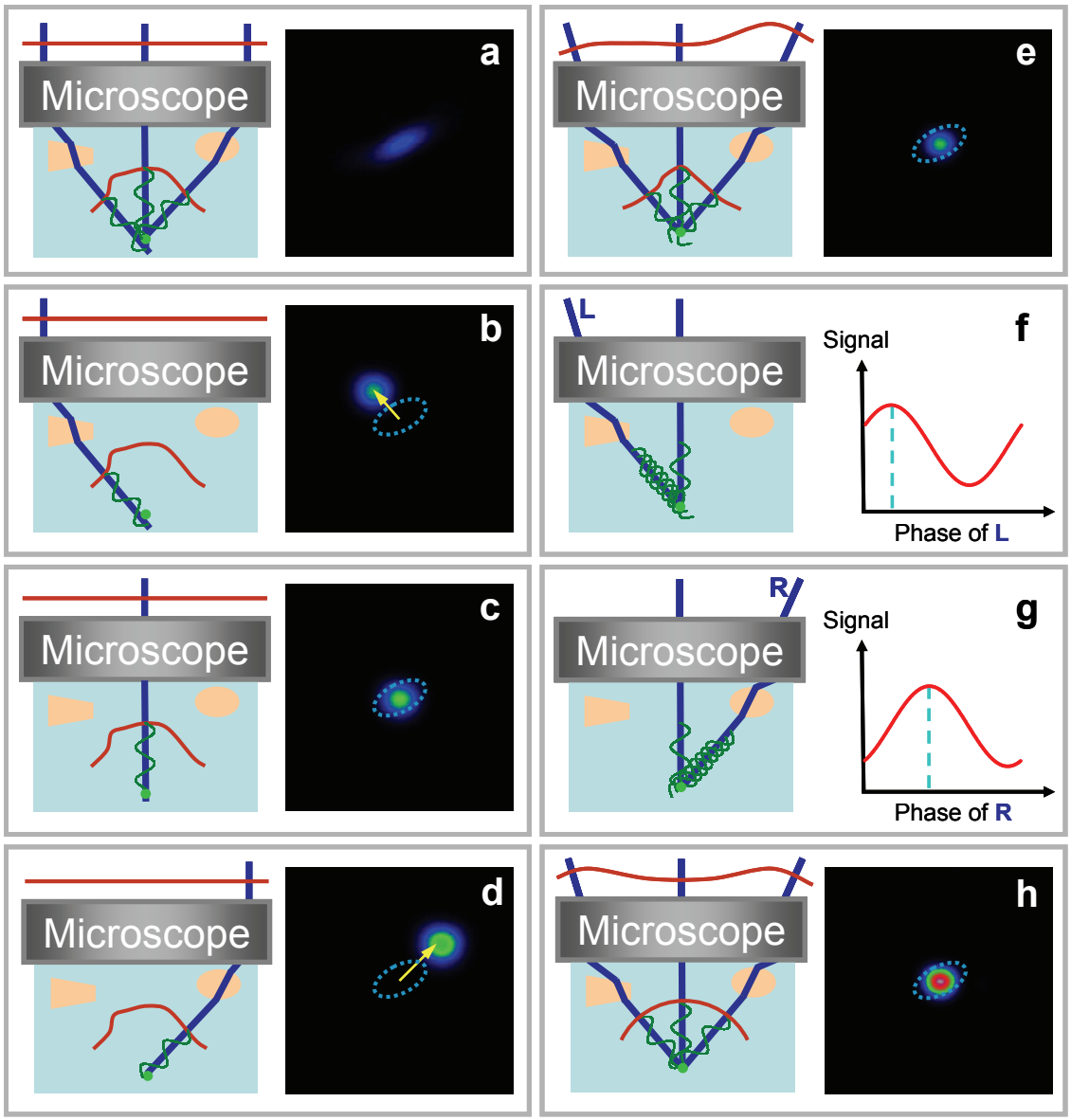



**Supplementary Figure 1.** (**a**) The essential components of our AO two-photon fluorescence microscope. (**b**) Detailed components of the microscope: Ti:Sapphire laser; electro-optical modulator (EOM); 2× beam expander (BE), beam pickoff (BP) that reflects ~3% of the light into a photodiode (PD); neutral-density filter wheel (FW); 2× beam reducer on motorized translation stage (BR); pair of mirrors mounted on independent motorized stages for two-axis laser positioning (LP); X galvanometer (X), two 30mm focal-length telecentric f–θ lenses (F1); Y galvanometer (Y); 30mm (F1) and 150mm (F5) focal-length telecentric f–θ lenses; spatial light modulator (SLM), pair of telecentric f–θ relay lenses (two 150mm focal-length lenses (F5) for Zeiss objective configuration, one 120mm (F4) and one 240mm (F8) lens for Nikon objective configuration); field stop (FS) at an intermediate image plane between them; protected-silver mirror (M1); dichroic long-pass beamsplitter (TPE BS) to separate excitation light from fluorescence;, Zeiss or Nikon water dipping objective (Obj) mounted on a Z-piezo stage (ZP); sample (S); fluorescence collimating lens (L1); dichroic fluorescence beamsplitter (FLR BS) to separate green and red fluorescence; two lenses (L2, L3) to focus fluorescence through two filter sets (FL1, FL2) onto two photomultiplier tubes (PMT).



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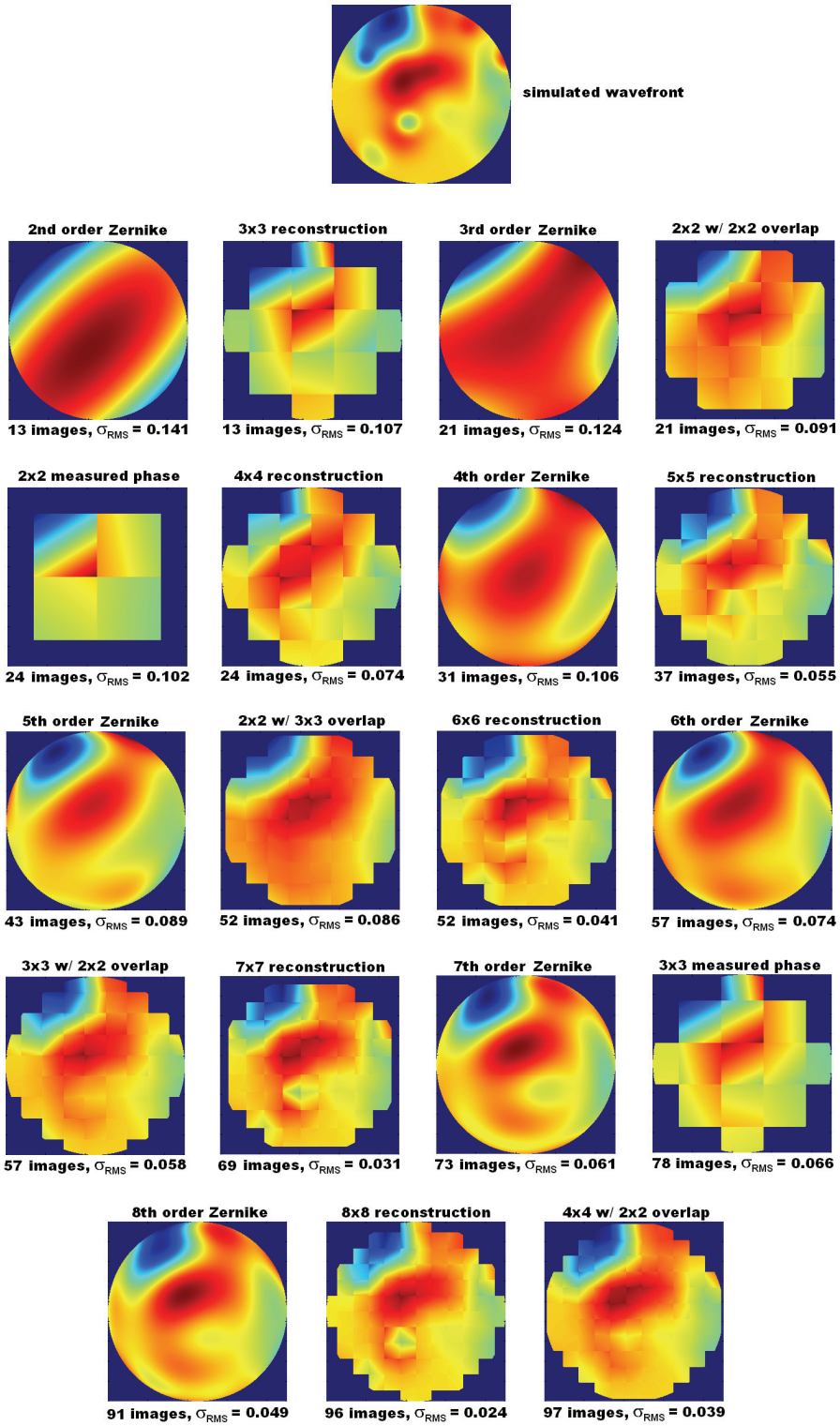
**Supplementary Figure 2** Step-by-step description of our pupil-segmentation based AO algorithm.




**Supplementary Figure 2.** Sketches illustrating our AO algorithm using three independent subregions, creating three beamlets (blue rays). (**a**) Aberrated wavefront (red) due to refractive index inhomogenieites (orange) leads to an aberrated image of a reference bead. (**b,c,d**) Images acquired with the left, center, and right subregions, respectively, permit the tilt of each beamlet to be measured from the displacement of the bead. (**e**) Beamlets intersect at a common point, after appropriate compensatory tilts are applied at the SLM. (**f**) Interference of left beamlet with central reference beamlet, at several phase offsets (green sinsusoids) applied to the former, determines the optimal phase offset (dashed aqua line); (**g**) Same procedure applied to the right beamlet. (**h**) Final corrected wavefront (red) and recovered diffraction-limted focus. An example with actual data showing the images and SLM patterns during the AO correction with $N = 36$ subregions is given in **Supplementary Movie 1**. If phase reconstruction is used, the phase measurement steps (**f** and **g**) are skipped, and the optimal phase offset in each subregion is determined algorithmically.

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**Supplementary Figure 3** Simulations comparing modal and zonal representations of a complex aberration pattern, the latter obtained using different variants of our AO algorithm.





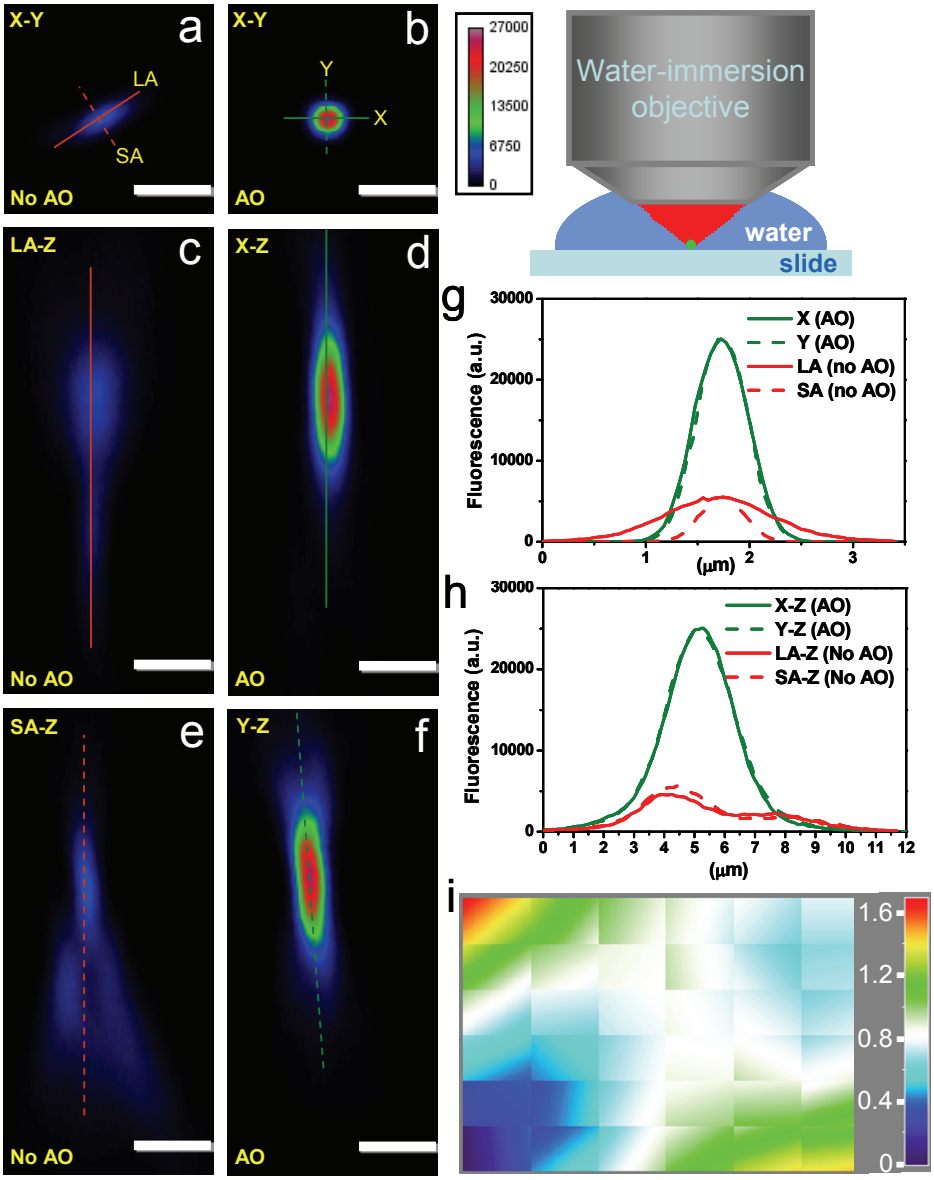

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**Supplementary Figure 3.** Modal and zonal representations of a complex aberration pattern (top image). The representations are ordered by the number of images required for AO correction, using the method listed at the top of each. For modal representations (e.g., "$Z^{th}$ order Zernike"), $2N+1$ images are needed to correct for all $N$ Zernike modes through order $Z$ (Ref. 25). Zonal representations were estimated using different variants of our pupil segmentation algorithm (see Results). Measured phase requires $6N$ images for $N$ subregions, assuming one for beam deflection measurement and five for phase. Phase reconstruction requires only $N$ images for $N$ beam deflection measurements. Higher density sampling is achieved with stepped overlap, but at the cost of more images. The root mean square error $\sigma_{RMS}$ between each representation and the original simulated wavefront, normalized to the peak-to-peak amplitude of the latter, is listed for each scenario, and represents the average from 25 simulated wavefronts. For a similar number of images required, zonal representations of the wavefront obtained via phase reconstruction with either independent or stepped overlapping masks always describe complex aberration patterns more accurately than the modal representation.



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**Supplementary Figure 4** Correction of system aberration.

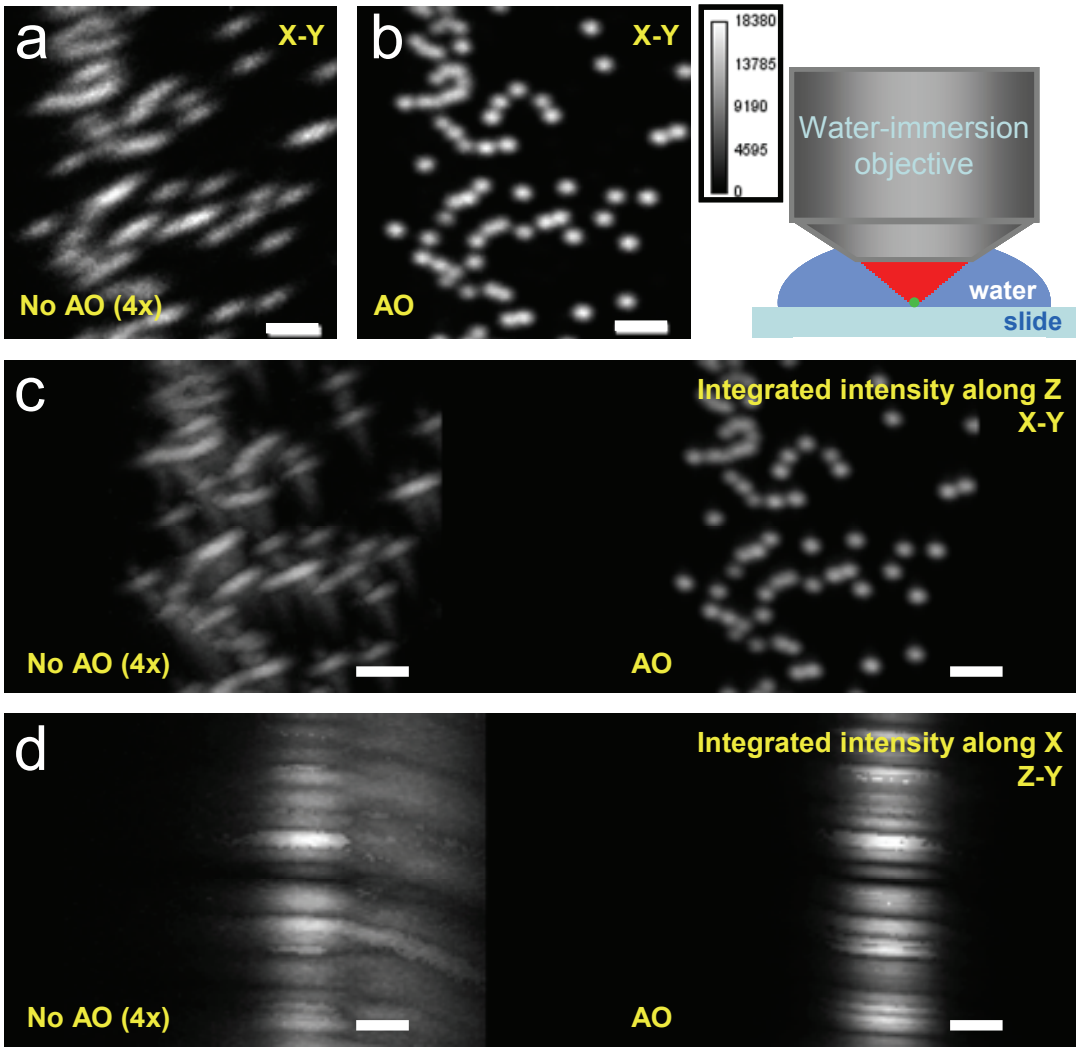



**Supplementary Figure 4.** Images of a 500 nm fluorescent bead immersed in the design medium of water (inset, upper right), both before (**a,c,e**) and after (**b,d,f**) AO correction with $N = 36$ independent subregions and measured phase (see Results). In both cases, three orthogonal image planes were selected: X–Y, and either LA-Z, SA-Z or X-Z, Y-Z, where Z denotes the axial direction, and LA and SA denote the long and short axes, respectively, of the bead image before correction. (**g**) Comparative intensity profiles along lines drawn in the X-Y lateral plane (red and green in **a**,**b**). (**h**) Similar profiles along lines drawn in the axial planes (red and green lines in **c**,**d**,**e**,**f**). (**i**) The final corrective wavefront for system aberrations in our microscope when using a Zeiss 20x, 1.0NA objective, in units of wavelength λ (λ = 850nm). Scale bar: 2 μm.



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**Supplementary Figure 5** Application of system correction obtained from one bead to a field of beads.

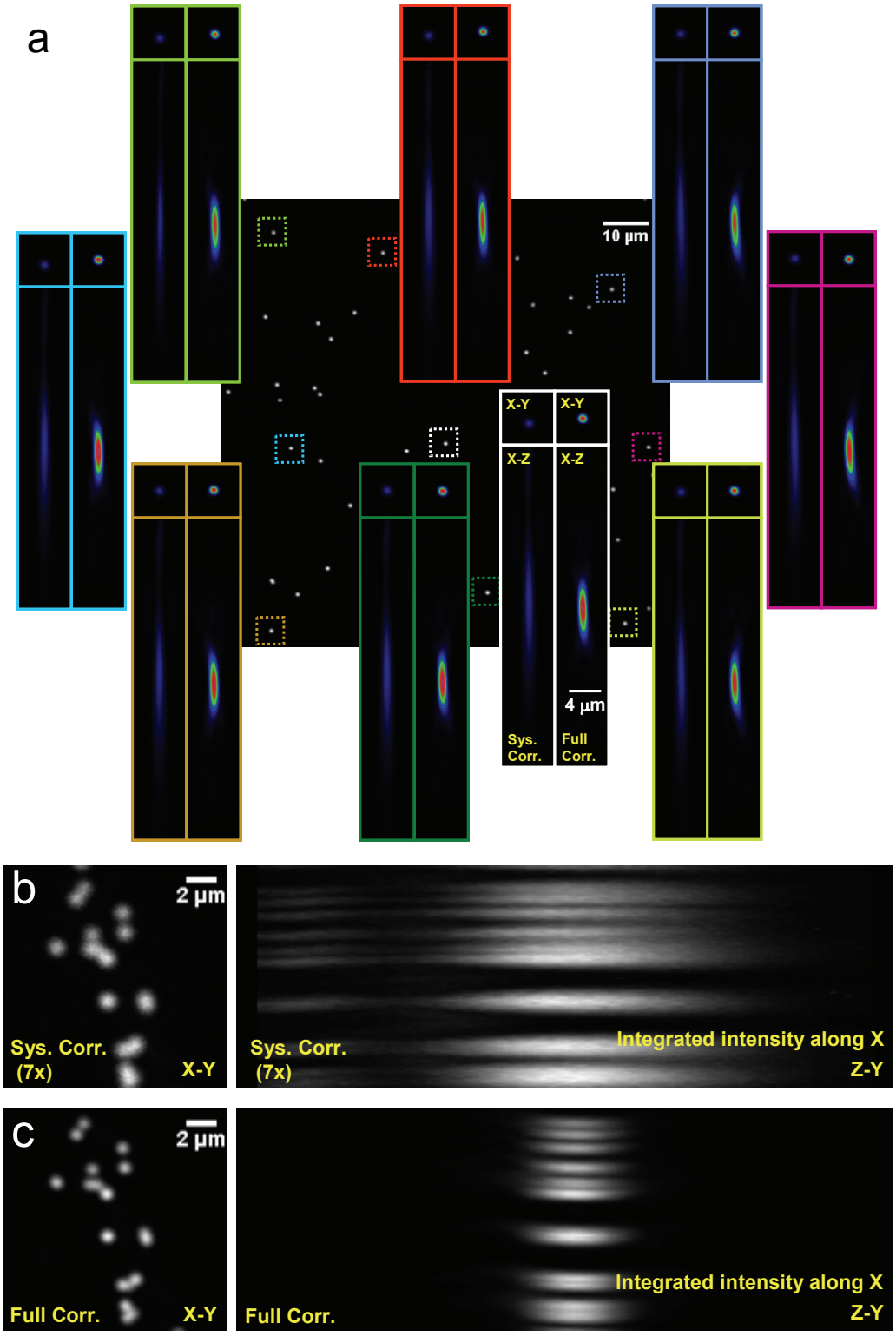



**Supplementary Figure 5.** Applicability of AO correction for system aberration at one point over a larger field of view. (**a**) Lateral image of a field of 500 nm diameter fluorescent beads before AO correction. (**b**) Image of the same beads, after applying the system correction determined from the central bead. (**c**) Lateral (X-Y) integrated intensity projection obtained by summing the entire 3D image stack along the axial Z direction. (**d**) Axial Z–Y intensity projection integrated along the lateral X direction. Display gain of the left, uncorrected images in **a-c** is increased 4× in (**c**,**d**) to improve visibility. Rotating integrated intensity projections distilled from the uncorrected and corrected 3D image stacks are shown in **Supplementary Movie 2**. Scale bar: 2 μm.



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**Supplementary Figure 6** Aberration correction for a field of beads in air observed with an objective designed for water-immersion.





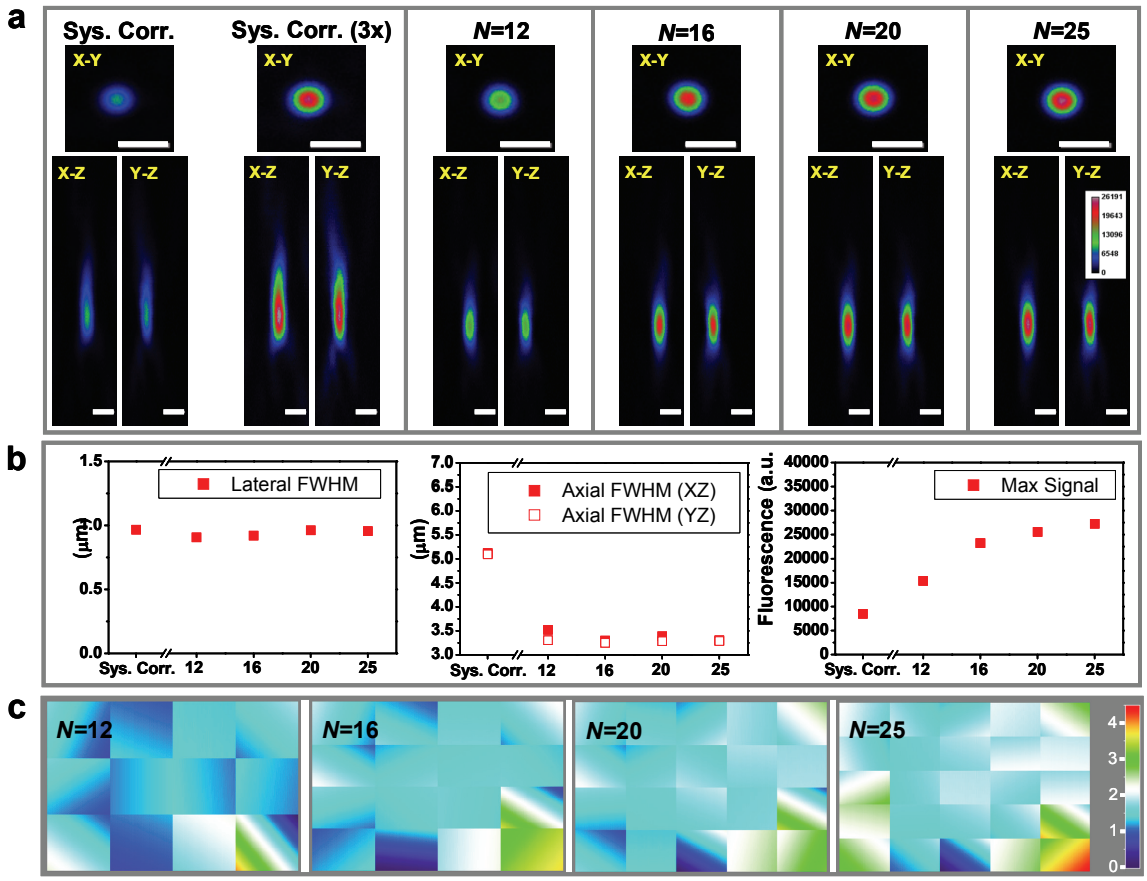

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**Supplementary Figure 6.** Aberration correction for two different fields (**a** and **b**,**c**) of 500 nm diameter fluorescent beads in air, observed with an objective designed for water-immersion. (**a**) Lateral (X-Y) image after correction of a 96 × 96 μm field of beads. Insets show lateral (X-Y) and axial (X-Z) images of individual beads at different field positions, comparing system aberration correction only with full AO correction, the latter obtained from one bead near the center of the field of view. (**b**) Lateral (X-Y) image of another field of beads acquired with system aberration correction only, and the associated axial Z-Y intensity projection integrated along the lateral X direction. (**c**) Corresponding images of the same beads, using full AO correction. The display gain of the system corrected images is increased 7× in (**b)** to improve visibility. Rotating integrated intensity projections distilled from the 3D image stacks for the system correction and full AO correction are shown in **Supplementary Movie 3**.



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**Supplementary Figure 7** Quality of aberration correction versus the number of corrective subregions *N*.

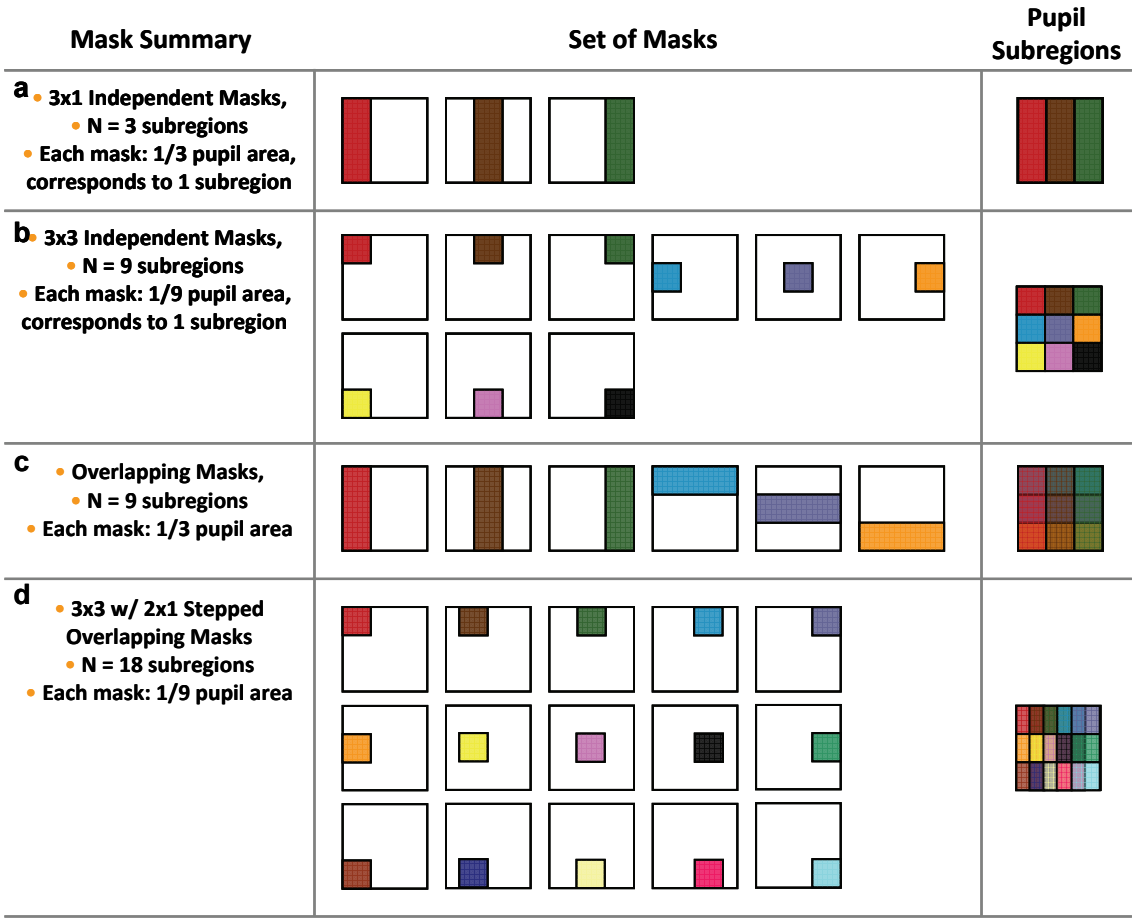



**Supplementary Figure 7.** (**a**) Lateral (top) and axial (bottom) images of a 1 μm bead as viewed through 250 μm thick fixed mouse brain slice under different correction conditions as noted. (**b**) Lateral and axial full width at half maximum and peak signal for each of the correction conditions in (**a**). (**c**) The final corrective wavefront in units of wavelength λ (λ = 850nm), after subtraction of system aberrations, for the cases $N = 12$, $N = 16$, $N = 20$, and $N = 25$, obtained using the independent mask algorithm with direct phase measurement. Scale bar: 2 μm.



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**Supplementary Figure 8** Comparison of the independent, overlapping, and stepped overlapping mask approaches to AO correction.

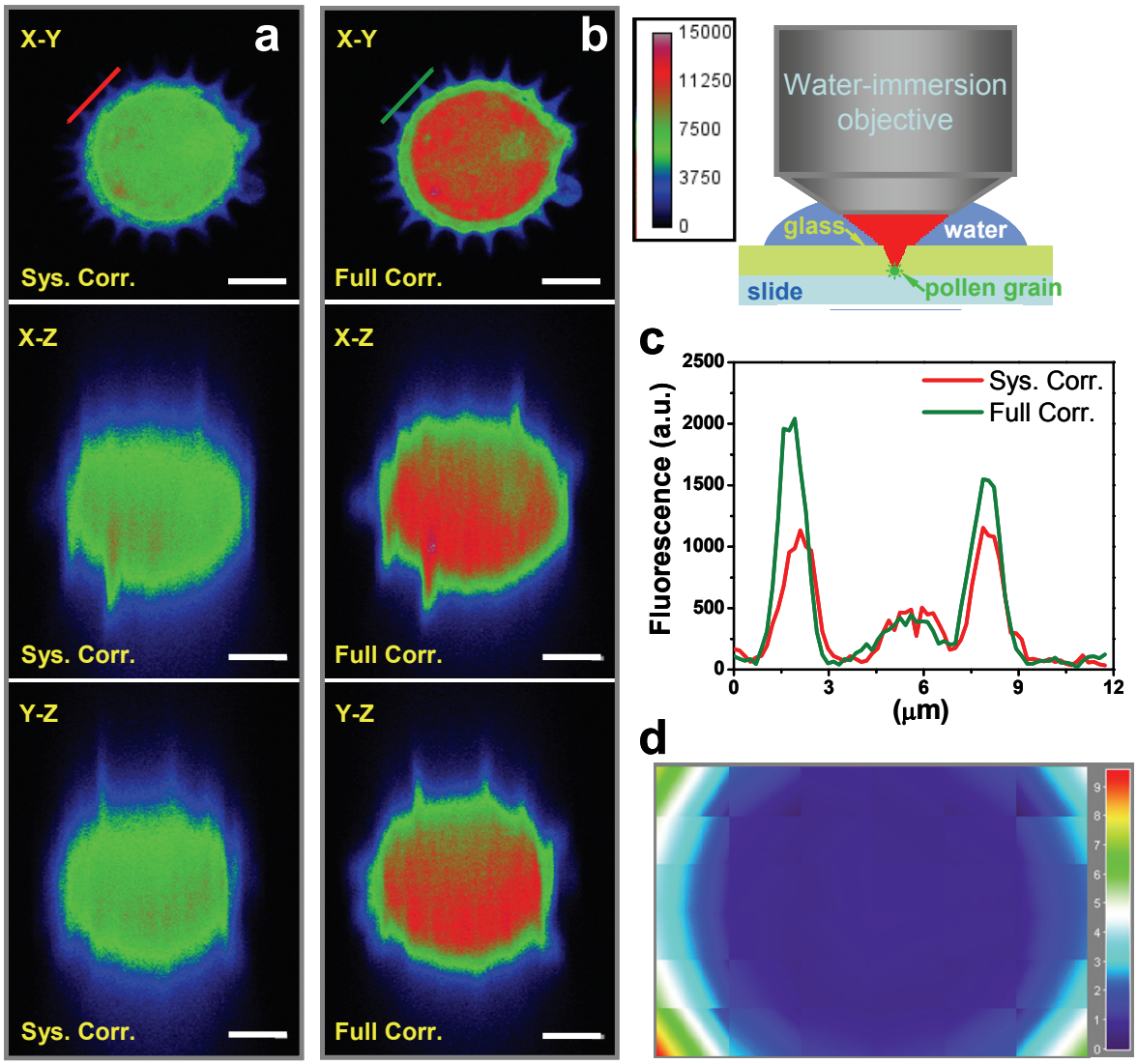



**Supplementary Figure 8.** Examples of the independent, overlapping, and stepped overlapping mask approaches to AO correction. The objective rear pupil is represented by the large square, and the colored rectangles represent the fraction of the pupil turned on at specific points during beam deflection measurement. (**a**) Independent mask approach: three non-overlapping masks, each covering 1/3 of the total pupil area, independently measure beam deflection in each of three pupil subregions. (**b**) Independent mask approach: nine non-overlapping masks, each covering 1/9 of the pupil area, independently measure beam deflection in each of nine pupil subregions. (**c**) Overlapping mask approach: six overlapping masks, each covering 1/3 of the pupil area, lead to unique wavefront estimations in each of nine pupil subregions. (**d**) Stepped overlapping mask approach: a mask covering 1/9 of the pupil area is translated in horizontal steps equal half the width of the mask, and in vertical steps equals to its height. Beam deflection is measured at each position. The example shown is denoted as 3×3 with 2×1 stepped overlapping masks. See **Online Methods** for detailed explanation.



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**Supplementary Figure 9** Aberration correction using centroid measurement without reference beads improves images of a fluorescent pollen grain under glass.

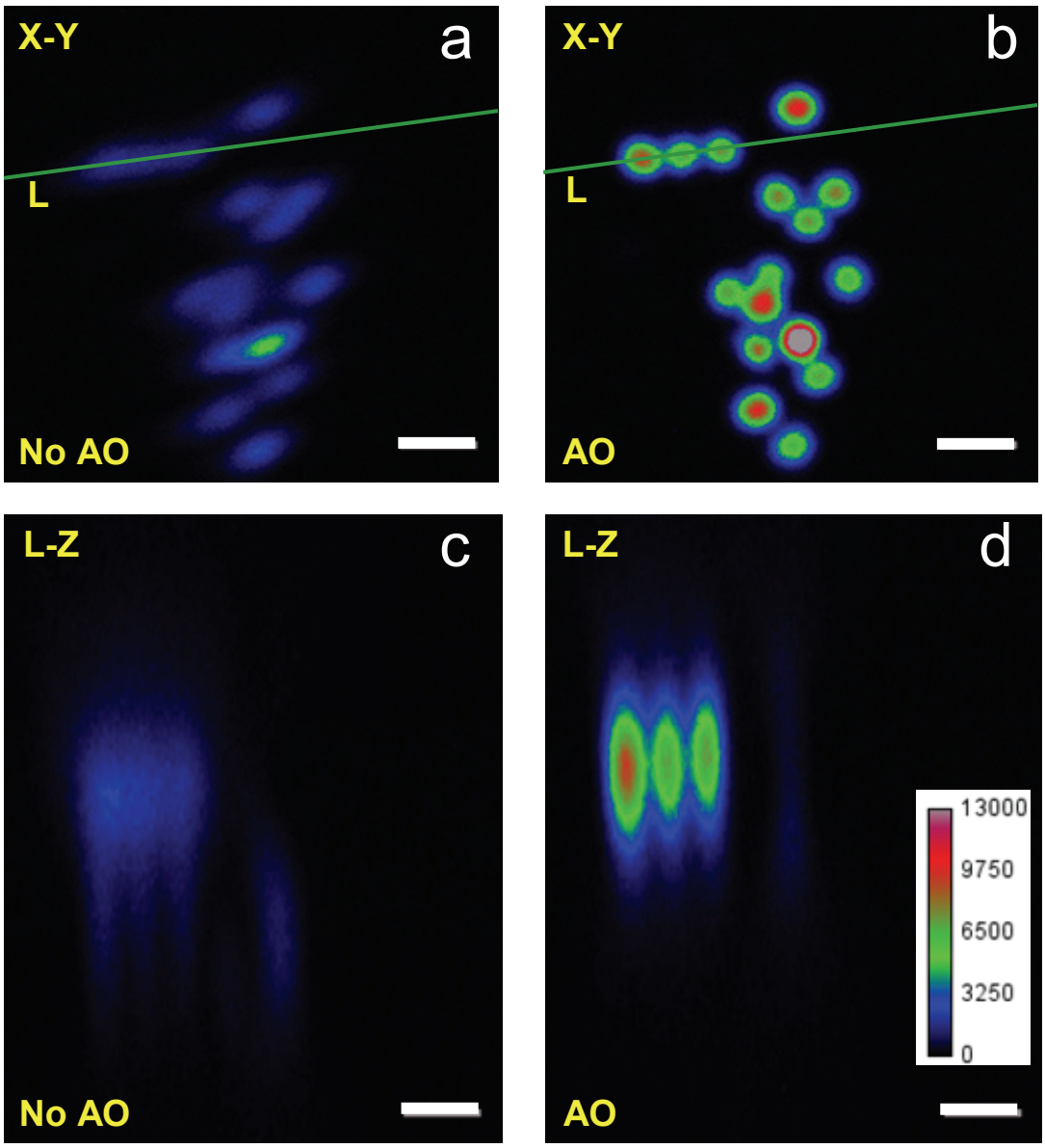



**Supplementary Figure 9**. Lateral and axial images of a fluorescent pollen grain under four aberration-producing cover glasses (inset), with system aberration correction only (**a**) and full AO correction (**b**). The centroid of the grain itself was used for AO correction, since no reference beads were added. (**c**) Intensity profiles along the colored lines in (**a**,**b**). (**d**) Final corrective wavefront in units of wavelength λ (λ = 850nm), after subtraction of system aberrations, using the independent algorithm with $N$ = 36 subregions and direct phase measurement. Scale bar: 10 μm.



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**Supplementary Figure 10** Correction for system aberration using image correlation on a collection of beads in water.

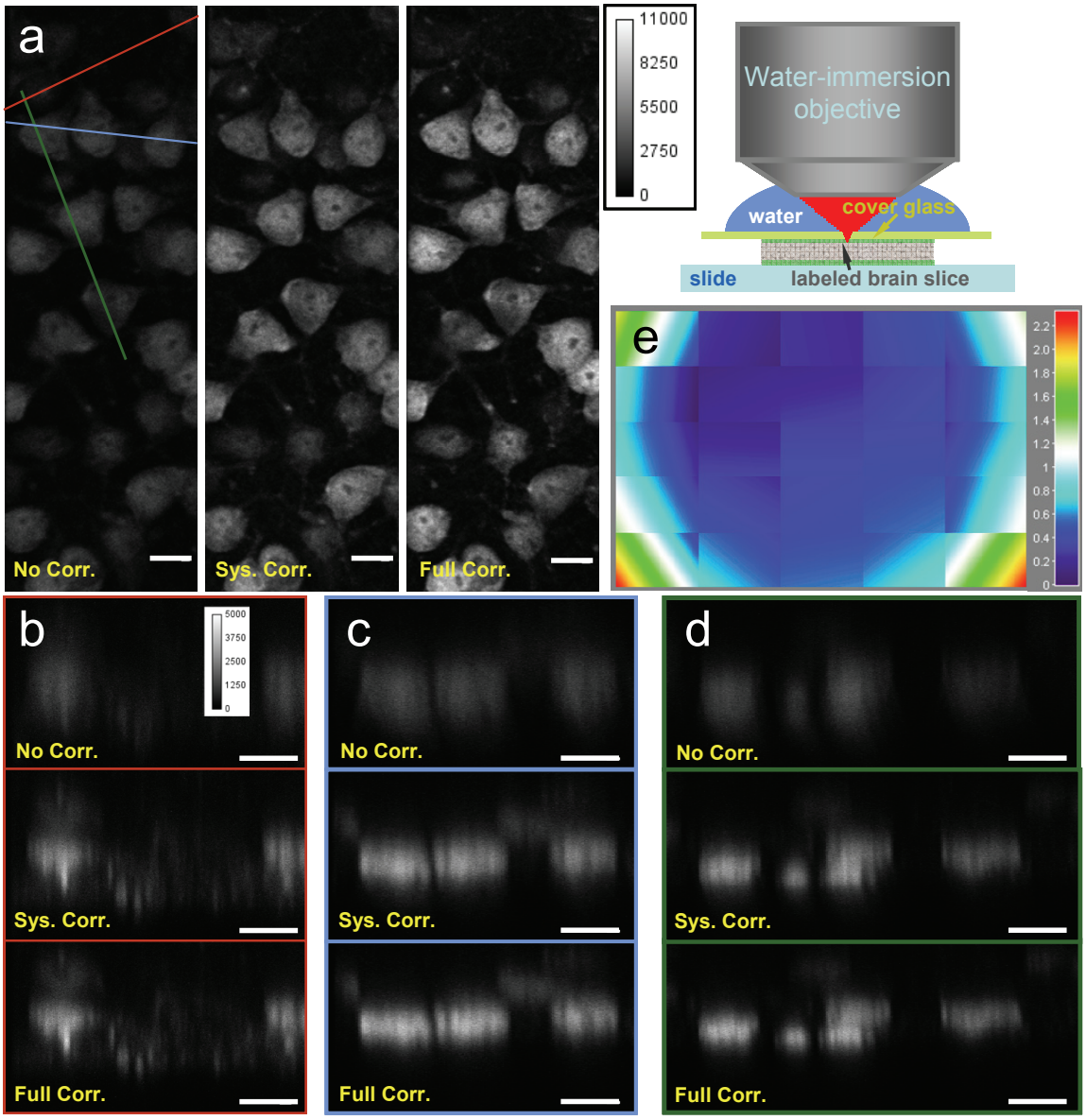



**Supplementary Figure 10.** **(a)** Lateral image of a field of 500 nm and 1 μm fluorescent beads in water, obtained with no aberration correction. (**b**) Same field of beads, after correction for microscope system aberrations, as determined using image correlation. (**c**) Image without correction, in the axial plane containing the green line in (**a**). (**d**) Aberration corrected image, in the same axial plane. Scale bar: 2 μm.



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**Supplementary Figure 11** Aberration correction at the top surface of an antibody-labeled 300 μm thick fixed mouse brain slice.

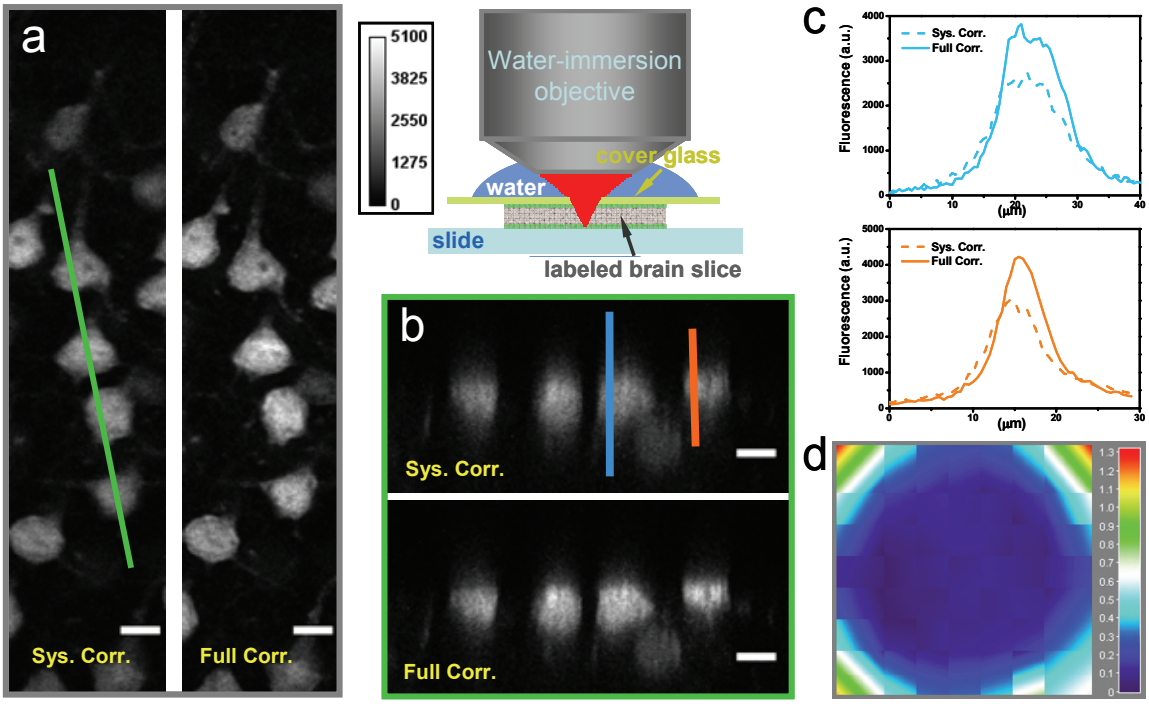



**Supplementary Figure 11.** Aberration correction at the top surface of a glass covered, antibody-labeled 300 μm thick fixed mouse brain slice (inset, top right), obtained using image correlation for beam deflection measurements. (**a**) Lateral images of a field of neurons acquired with, from left to right, no correction, system correction only, and full AO correction. (**b,c,d**) Images in the axial planes defined by the red, blue, and green lines in **a**, respectively. (**e**) The final corrective wavefront in units of wavelength λ (λ = 850nm), after subtraction of system aberrations, using the independent mask algorithm with $N = 25$ subregions and direct phase measurement. Scale bar: 10 μm.



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**Supplementary Figure 12** Aberration correction at the bottom surface of an antibody-labeled 300 µm thick fixed mouse brain slice using image correlation and phase reconstruction.

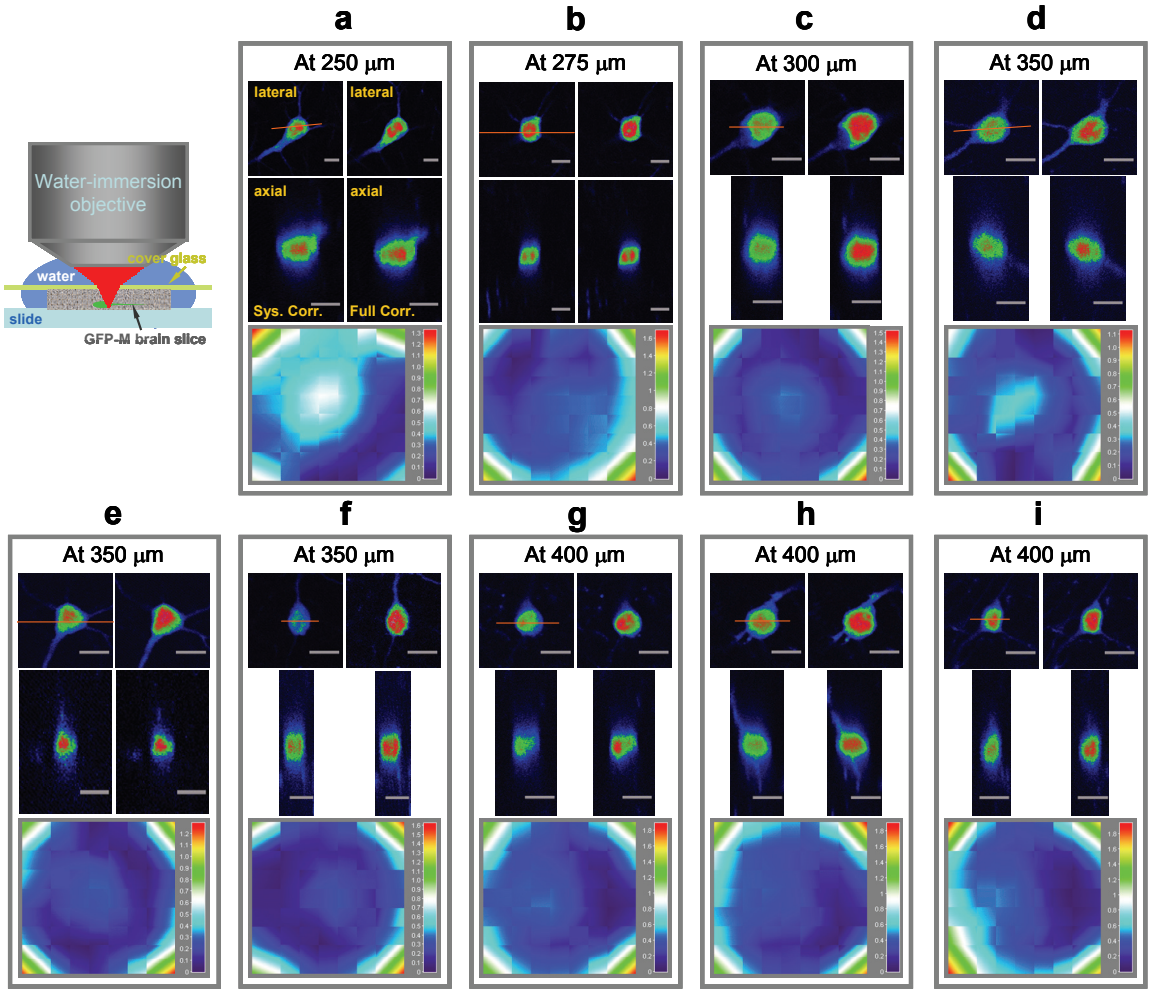



**Supplementary Figure 12.** Aberration correction at the bottom of a glass covered, antibody-labeled 300 µm thick fixed mouse brain slice (inset, middle), obtained by image correlation and phase reconstruction. (**a**) Lateral images of a field of neurons acquired with, from left to right, system correction only and full AO correction. (**b**) Images in the axial planes defined by the green line in **a**. (**c**) Comparative intensity profiles along the blue and orange lines in (**b**). (**d**) The final corrective wavefront in units of wavelength λ (λ = 850 nm), minus system aberrations, using the 4x4 with 2x2 stepped overlapping mask algorithm and phase reconstruction. Scale bar: 10 µm.



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**Supplementary Figure 13** Aberration correction of GFP-labeled neurons at different depths below the surface of fixed mouse brain slices using image correlation and phase reconstruction.

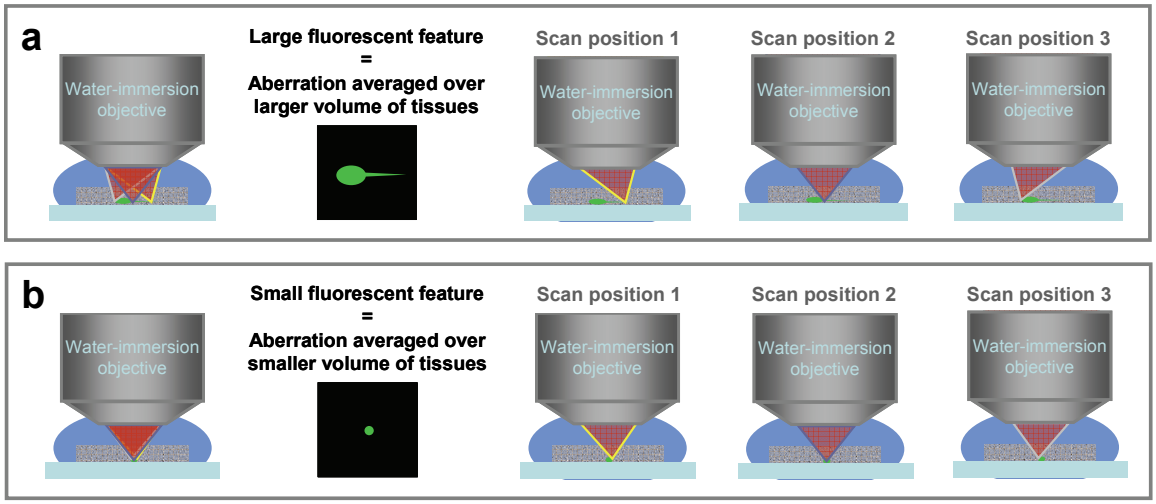



**Supplementary Figure 13.** Aberration correction of GFP-labeled neurons at (a) 250 μm, (b) 275 μm, (c) 300 μm, (d) 350 μm, (e) 350 μm, (f) 350 μm, (g) 400 μm, (h) 400 μm, and (i) 400 μm below the surface of fixed mouse brain slices, as obtained using image correlation. At each depth, lateral and axial images of the target neuron are shown with system correction only and full AO correction, as well as the corrective wavefront in units of wavelength λ (λ = 900nm) (after subtraction of system aberrations). The 4×4 with 2×2 stepped overlapping mask algorithm and phase reconstruction was used in each case. Scale bar: 10 μm.



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**Supplementary Figure 14** Sketches showing the relationship between size of the field over which AO correction is performed to the volume of tissue over which the correction is averaged.

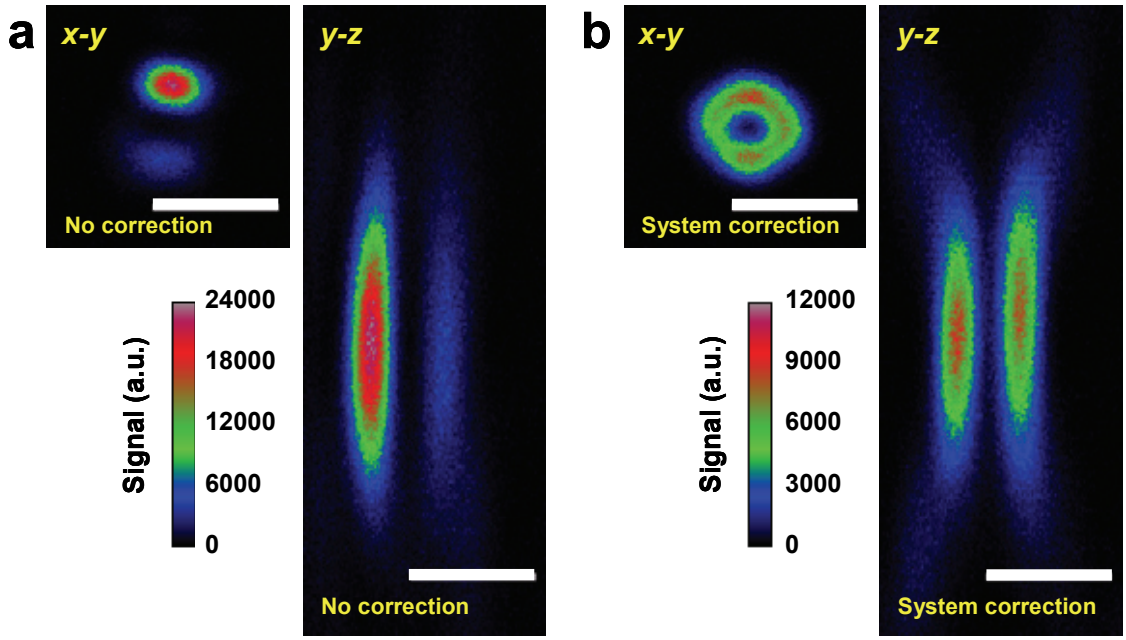



**Supplementary Figure 14.** (**a**) When fluorescent features spread over a large field of view are used for AO correction, the excitation beam probes a large volume of biological tissue during correction: the distinct locations of the excitation light inside the tissue at scan positions 1, 2, and 3 show that the measured aberration will represent an average over this large volume. (**b**) When AO correction is performed over a small field of view, the excitation light probes the same, smaller tissue volume at all scan positions, leading to a correction which is more accurate locally, but may be less accurate elsewhere than the wider, field averaged correction.



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**Supplementary Figure 15** Effects of system aberration and AO correction of same on the generation of an annular PSF such as is used in stimulated emission depletion microscopy.

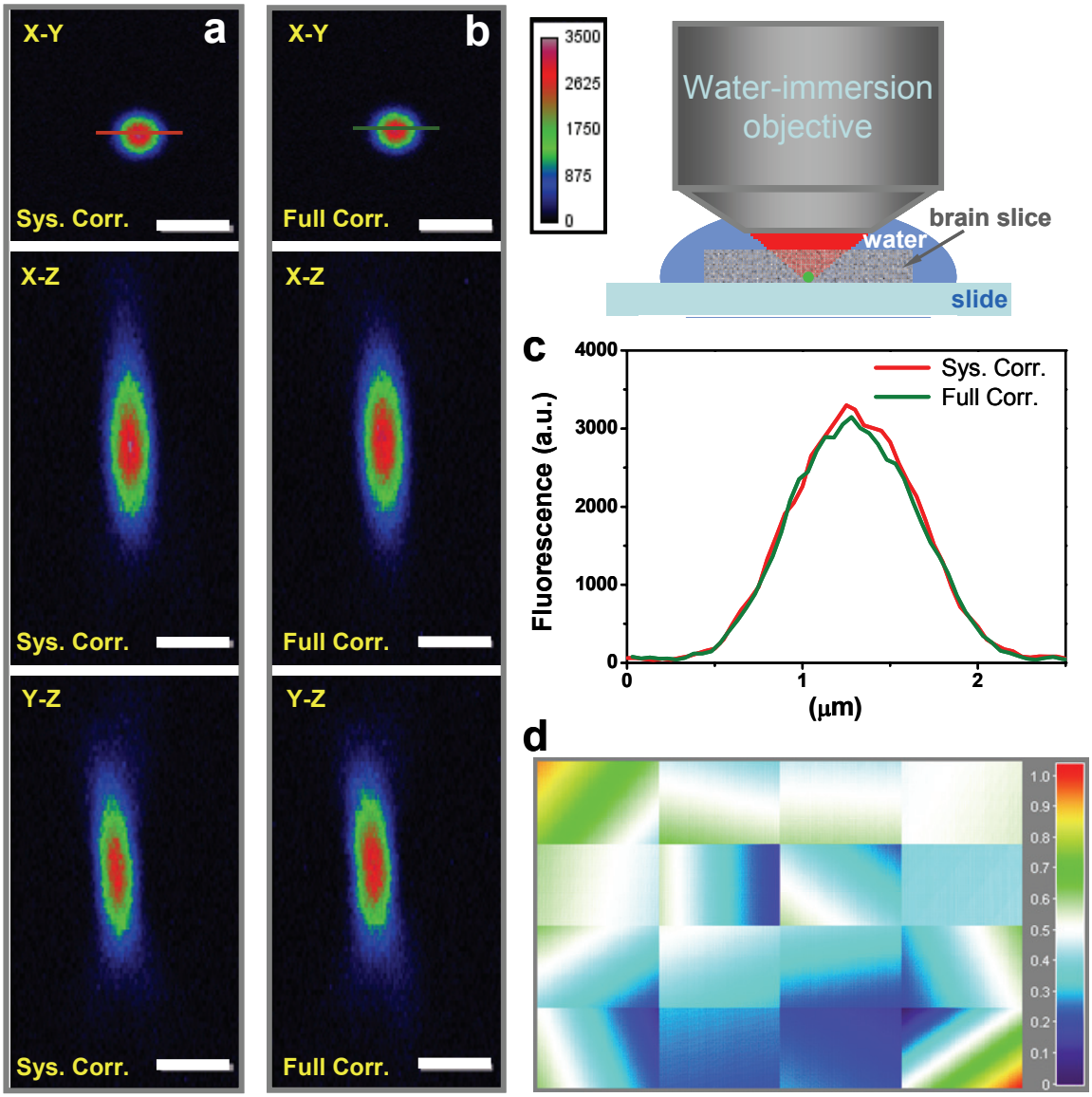



**Supplementary Figure 15.** Generation of an annular PSF (**a**) without and (**b**) with correction for system aberration. Lateral (*x*-*y*) and axial (*y*-*z*) images of a single 500 μm diameter bead were taken with a helical phase ramp from 0 to 2π on the SLM. With no aberration correction (**a**), the resulting PSF was severely distorted; with system aberrations corrected (**b**), the desired annular PSF with minimal central intensity was obtained. Scale bar: 2 μm.



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**Supplementary Figure 16** An example of aberration correction through a 250 μm brain slice that did not detect significant aberration.

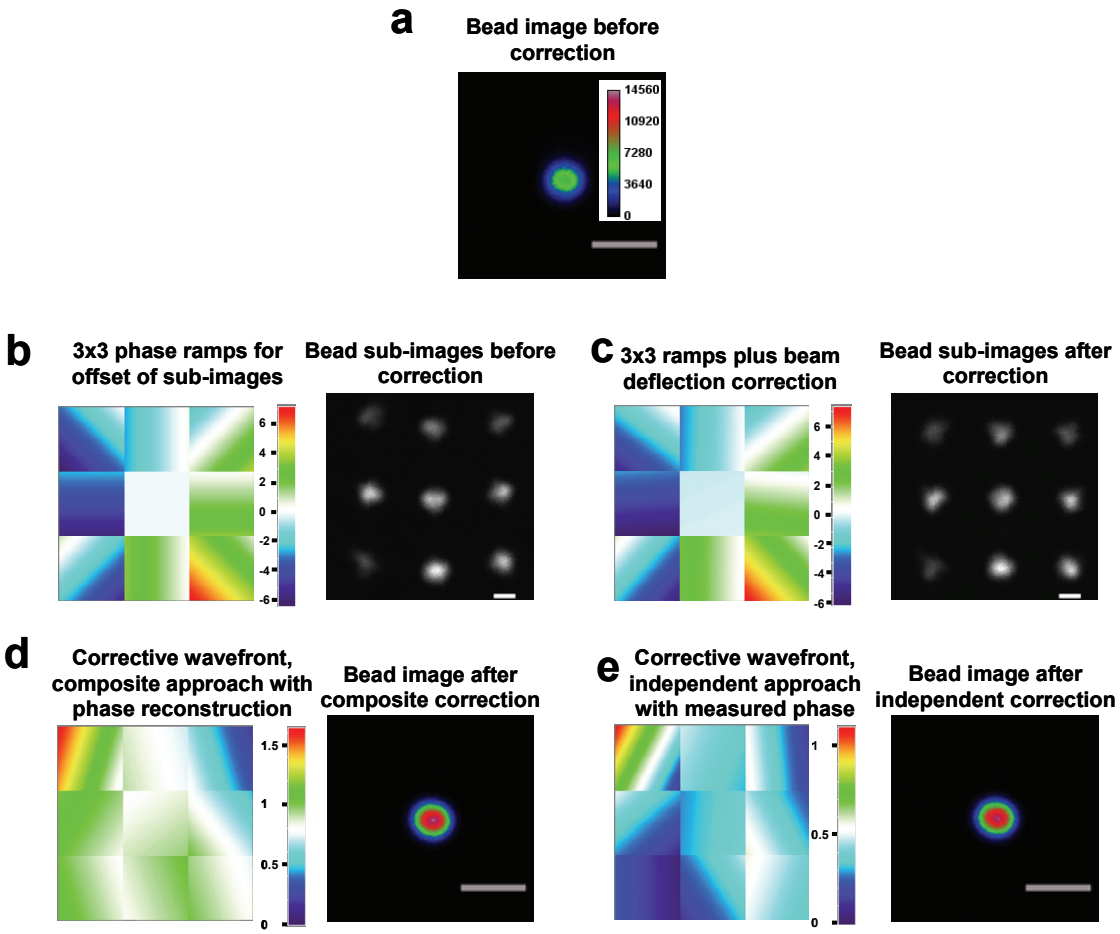



**Supplementary Figure 16.** Lateral and axial images of a single 1 μm diameter bead under a 250 μm fixed brain slice (inset) with system correction only (**a**) and full AO correction (**b**). (**c**) Intensity profiles along the colored lines in (**a**,**b**). (**d**) Final corrective wavefront in units of wavelength λ (λ = 850nm), after subtraction of system aberrations, using the independent algorithm with $N$ = 12 subregions and direct phase measurement. Scale bar: 2 μm.



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**Supplementary Figure 17** AO correction of system aberration using only a single composite image.

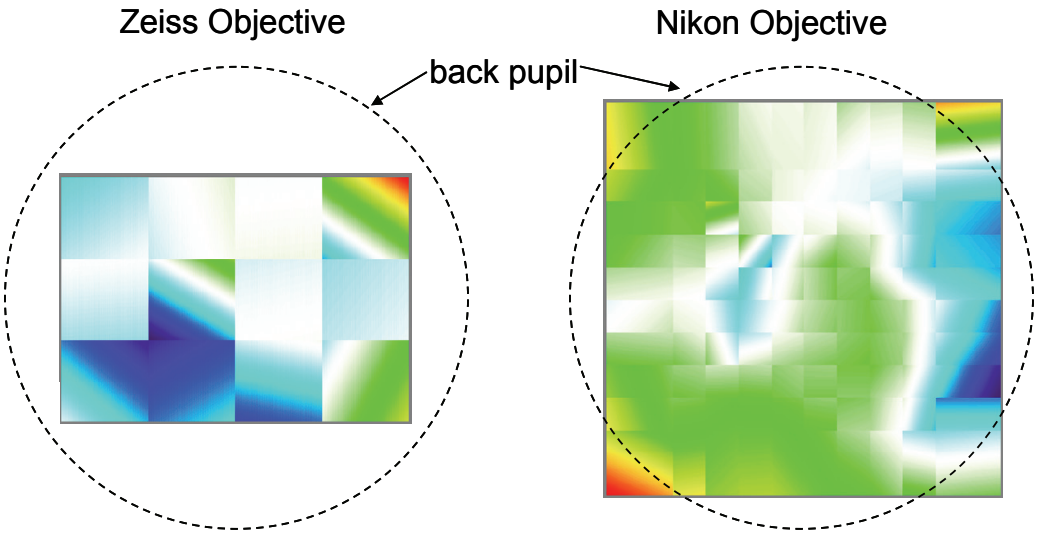



**Supplementary Figure 17.** AO correction for system aberration using only a single composite image. (**a**) Image of a 1 μm bead when no correction for system aberration was applied to the SLM. (**b**) After simultaneously applying different phase ramps to each of the 3×3 subregions of the SLM, nine spatially offset sub-images of the same bead were seen. The deviations from the ideal 3×3 array that would exist in the absence of aberration indicate the aberration-induced tilt error that exists in each subregion. (**c**) Correcting these tilt errors causes the bead sub-images to become aligned with the ideal 3×3 array. (**d**) The final corrective wavefront obtained using the tilt error data and phase reconstruction, and the resulting corrected bead image, both of which were similar to those obtained by 3×3 independent masks and direct phase measurement (**e**). Scale bar: 2 μm.



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**Supplementary Figure 18** Dimensions of the SLM relative to the back apertures of the Zeiss and Nikon objectives.




**Supplementary Figure 18.** The active area of the SLM relative to the back apertures (dashed circles) of the Zeiss (left) and Nikon (right) objectives.



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